PHISHING DETECTION & AWARENESS REPORT

Email Analysis · Employee Security Education

Prepared by	Future Interns — Cyber Security Task 2
Report Date	15 April 2026
Target Audience	Internal Security Awareness
Assessment Type	Phishing Email Analysis & Risk Classification
Tools Used	Phishtool, Email Header Analysis, Google Admin Toolbox,
Classification	MXToolbox
Author	For Training Purposes Only
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1. Executive Summary

This report presents the results of a technical phishing email analysis conducted on a real-world sample (sample-1101) collected from a public phishing repository. The investigation identifies fraudulent indicators, technical email header inconsistencies, and social engineering tactics designed to harvest Microsoft user credentials.

Risk Identification Summary:

- Total Samples Analyzed: 1 (Microsoft Account Spoofing)
- Final Classification: CRITICAL
- Threat Type: Credential Harvesting (Phishing)
- Confidence Level: 100% — Multiple authentication failures confirmed

■RISK LEVEL: CRITICAL — PHISHING / CREDENTIAL HARVESTING

2. Technical Analysis (Sample-1101)

Source: github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1101.eml

2.1 Email Metadata

Displayed Sender	Microsoft account team
Actual Sender (From)	no-reply@access-accsecurity.com
Subject	Microsoft account unusual signin activity
Reply-To	solutionteamrecognizd03@gmail.com
Date Sent	Saturday, 12 August 2023 — 14:25:52 +0000
Sending IP	89.144.9.87 (hostname: nonkfrgr.co.uk)
Delivery Route	nonkfrgr.co.uk → Microsoft EOP → Exchange Labs → Outlook

2.2 Security Authentication Results

ThefollowingchecksweresperformedusingGoogle Admin Toolbox Messageheader Analyzer:

Protocol	Result	Security Impact
SPF	**NONE**	The sending IP (89.144.9.87) is not authorized by the claimed domain. High risk of spoofing.
DKIM	**NONE**	No digital signature present. Message content cannot be verified and may have been altered.
DMARC	**PERMERROR**	The sender domain fails all security policy checks. No valid DMARC record is configured.

Analyst Note: The combined failure of SPF, DKIM, and DMARC — alongside a Gmail Reply-To and a spoofed display name — constitutes a textbook phishing attempt with 100% confidence. The domain access-accsecurity.com is a classic typosquatting attack designed to visually mimic Microsoft's official domain.

2.3 Evidence — Email Header Screenshots

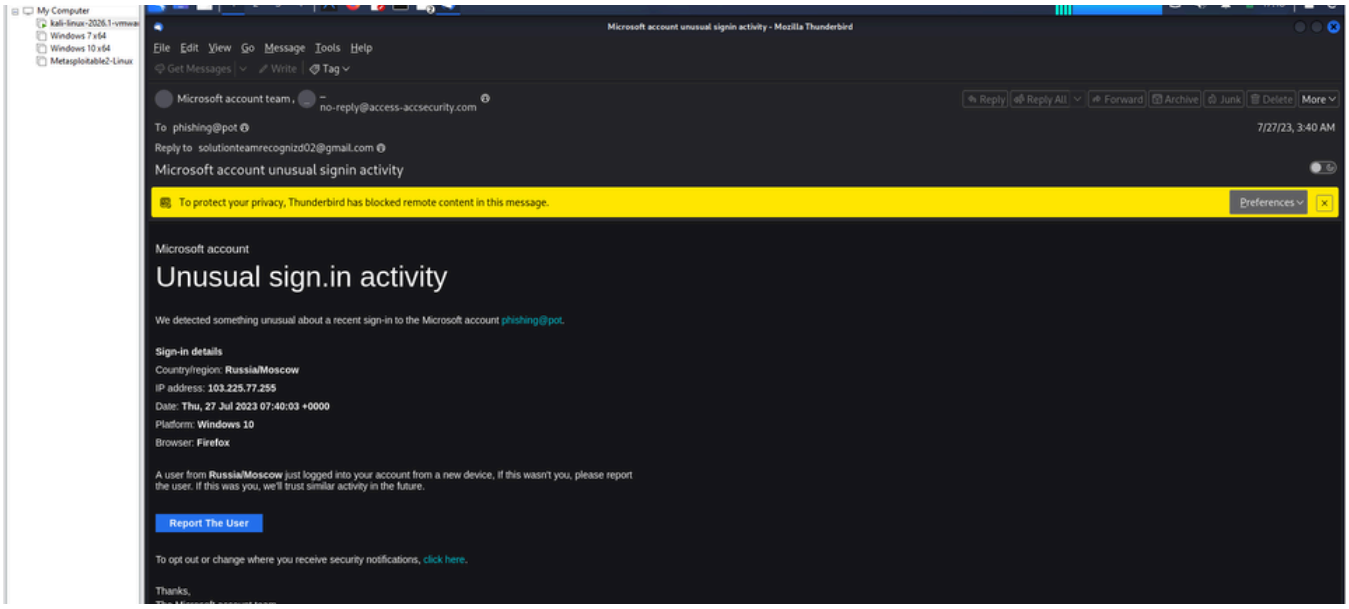

Figure 1: Email content

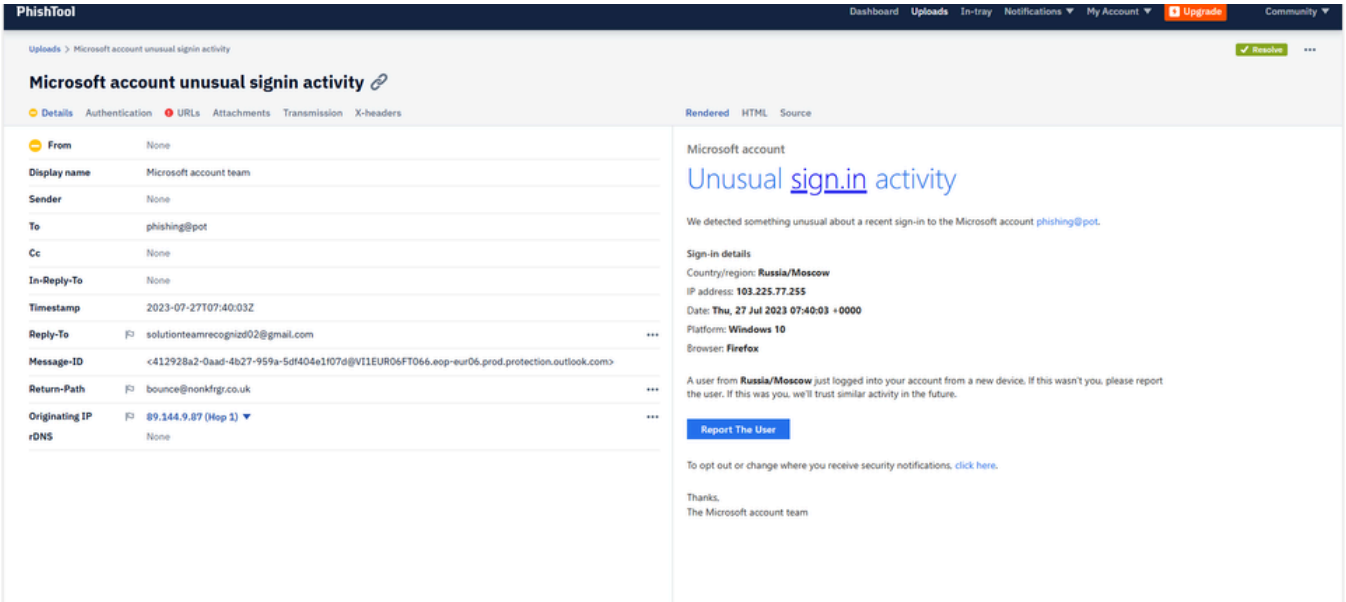

Figure 2: Upload email in phishtool

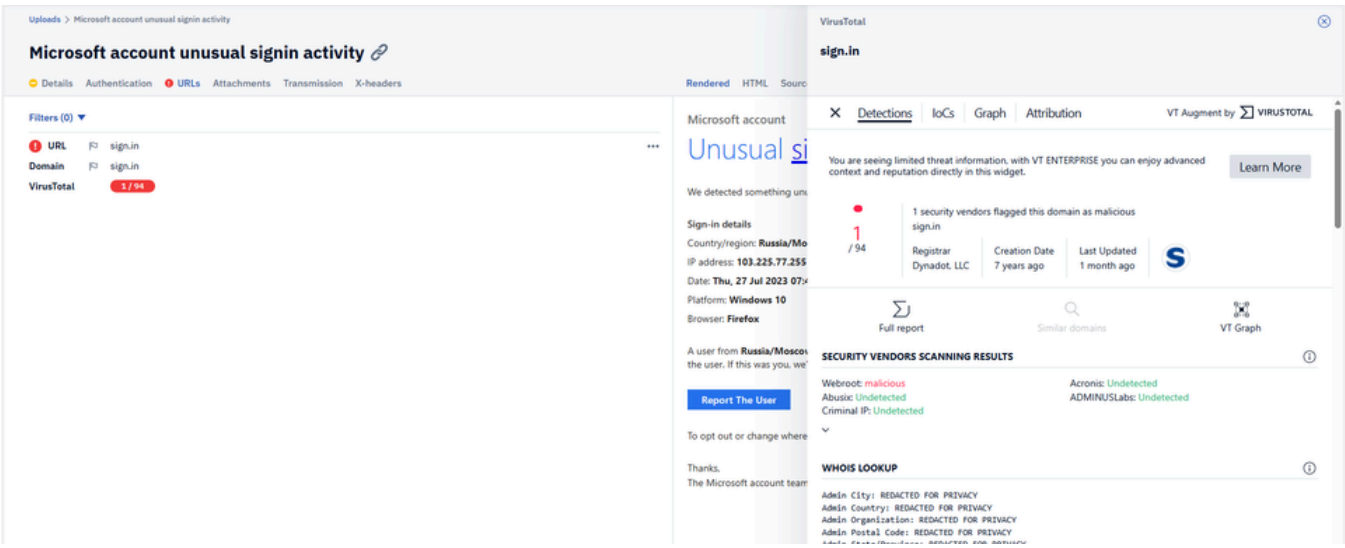

Figure 3: Malicious url analysis with virus total

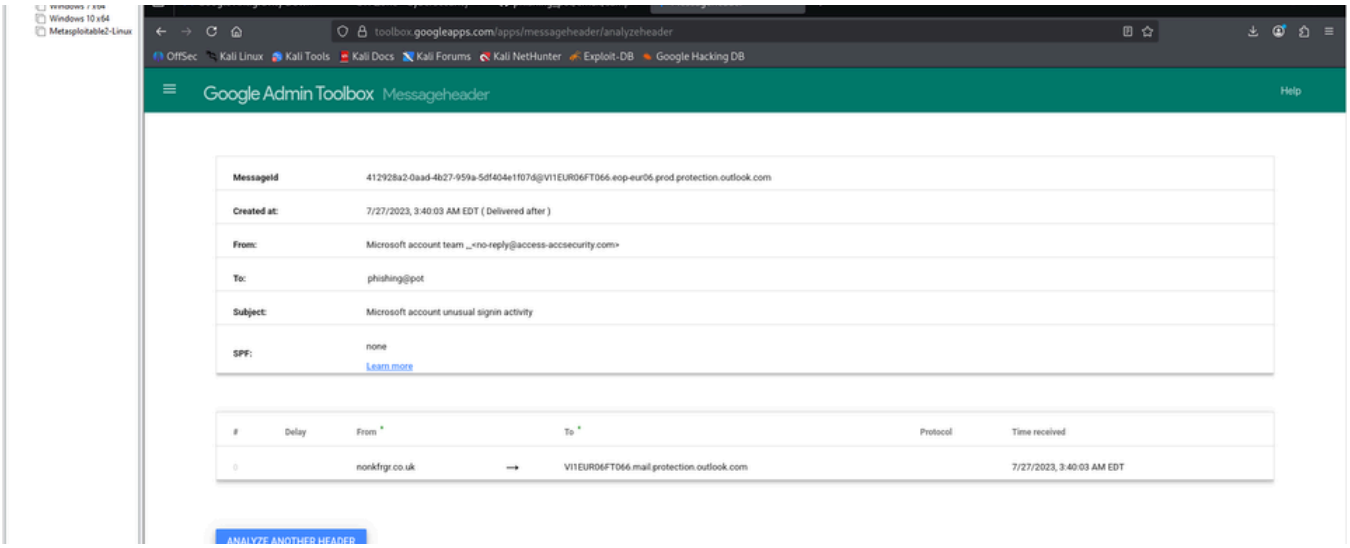

Figure 4: Google Admin Toolbox — SPF/DKIM/DMARC Results

3. Phishing Indicators Identified

3.1 Domain Mismatch (Display Name Spoofing)

Theemail displays thename**'Microsoftaccount team'** whilethe actual sender address is **no-reply@access-accsecurity.com**. This domain has no affiliation with Microsoft Corporation. Legitimate Microsoft security emails originate exclusively from @microsoft.com or @accountprotection.microsoft.com.

3.2 Fraudulent Reply-To Address

The Reply-To header(line 31ofthe raw source) directs any user response to **solutionteamrecognizd03@gmail.com** — a free personal Gmail address. No legitimate corporate security team uses a public Gmail account for security communications.

3.3 Unauthenticated Sending Infrastructure

The emailoriginatedfrom IP **89.144.9.87** (hostname:nonkfrgr.co.uk), which has no relationship to Microsoft infrastructure. The presence of Microsoft Exchange relays in the routing path is a deceptive technique to make the delivery chain appear legitimate.

3.4 Social Engineering Tactics

- Urgency & Fear:Subject implies immediate account risk ('unusual signin activity').
- Generic Salutation: Use of 'Dear User' instead of the recipient's real name.
- Authority Impersonation: Falsely claims to originate from the Microsoft security team.
- Call to Action: Pressures the victim to click a link and verify credentials urgently.

4. Awareness & Prevention Guide

This section is designed for non-technical employees. The goal is to empower every user to independently identify and report phishing emails before any damage occurs.

4.1 The 4 STOP Reflexes

S — Suspicion	Does this email create urgency, fear, or request unusual action?
T — Test the link	Hover over links (without clicking) to preview the real destination URL.
O — Origin check	Read the full email address — not just the displayed name.
P — Prudence	Never enter passwords or OTP codes on any site linked from an email.

4.2 Employee Best Practices

DO's — Actions employees should always take

Check the sender's full email address, not just the display name.

Hover over links to preview the real destination URL before clicking.

Report suspicious emails immediately using the 'Report Phishing' button.

Navigate to official websites directly via trusted browser bookmarks.

Contact your IT/Security team immediately if you are unsure about any email.

DON'Ts — Actions employees must always avoid

- ✘ Click on links in emails from unknown or unverified senders.
- ✘ Open attachments (especially .zip, .html, .exe) from external senders.
- ✘ Enter login credentials on any site reached from an email link.
- ✘ Reply to or interact with suspicious emails — do not test the attacker.
- ✘ Ignore or delete phishing emails without reporting them to IT first.

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5. Recommendations

5.1 For IT & Security Teams

- Block the domain access-accsecurity.com at the email gateway and firewall level immediately.
- Implement and enforce DMARC, DKIM, and SPF on all company-owned email domains.
- Configure email security gateways to quarantine or flag emails failing all three auth checks.
- Enable anti-spoofing rules to reject emails where the Reply-To differs from the From domain.

5.2 For Employees & End Users

- Attend regular phishing simulation training sessions organized by the security team.
- Remember: Microsoft, banks, and official services will NEVER use a Gmail address for security.
- When in doubt — do not click. Call the IT helpdesk directly to verify any suspicious email.

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6. Conclusion

The analyzed email (sample-1101) is a confirmed, high-confidence phishing attempt. It exploits display name spoofing, authentication failures (SPF/DKIM/DMARC), and psychological pressure tactics to steal Microsoft user credentials. This type of attack does not require a technical system weakness — it exploits human behavior.

By implementing the technical recommendations in Section 5 and educating employees with the awareness guidelines in Section 4, organizations can significantly reduce the risk of credential theft through phishing. **Security awareness is the most cost-effective defense against social engineering attacks.**